Econometrics Course Project: A Difference-in-Differences analysis
(…of, for example, a minimum wage increase)

In this group project, you will use data you construct to carry out an original analysis of the using a "difference-in-differences" approach, [1] and then write it up like an economics research article. It is up to you to develop the details of what outcome and event / policy you wish to examine, as well as the exact details of the estimation. Details below.

Timeline

Part	**Due Date**
1. Proposal	Oct 21
2. Balance Test/Preliminary data & methods section	Nov 2
3. Preliminary Results Section /rough draft	Nov 13
4. Final, complete draft (and log files)	Nov 17

1. **Proposal**. A short description stating what specific question you are asking, what variables you might want to use, and basic descriptive statistics. Details on Canvas.
   → See separate pages "Project Proposal Assignment/Getting Started" for details.
   - Include in this description what exact policy you will study, what your comparison group(s) will be, and the main outcome you will study. (It is possible to change/add to this later, but I want you to give it some thought before you start running your regressions and writing your paper).
   - A table of means of this outcome data and any key controls. This can be crude for now (essentially, it can be output of the "sum" command in STATA) – nicely formatted tables are not required until the final version of the paper.
2. **Preliminary "balance test" and Data / Methods section**. Creating a balance test, similar to Card & Krueger's Table 2, is a key first step in the analysis. Also, write up a description what exact data you are using, how you constructed your sample and variables, and what you're estimating will be useful in getting to a first draft.
   → See separate pages on "Data and Methods Section, Balance test" for more details.
3. **Preliminary project results section, with Tables/graphs**
   → See separate pages on "Results and Graphs" and "Creating Final Versions of Your Tables" for some thoughts
4. **Final, complete paper due!** Don't forget log files
   → See "Research Checklist" and the grading rubric for final checklist of things that you should try to have in the final version of your paper.
5. **Other things:**
   - I will try to give feedback as quickly as I can on your proposals, and other parts. The sooner you turn it in the better, because I can give feedback and …
   - **Turn in log files from any programs as well**.

[1] It is OK to study some other (that is non minimum wage) policy entirely, as long as its analysis can (a) fit reasonably well into a difference-in-differences framework and (b) you have a dataset in mind, including the one I recommend here, that can feasibly be used to do a sensible analysis of it.

How To Do It

The inspiration for this project is the 2021 Nobel-prize winning article on the impact of the 1992 minimum wage increase in NJ:

Card, David and Alan Krueger (1994) "Minimum Wages and Employment: A Case Study of the Fast-Food Industry in New Jersey and Pennsylvania." *American Economic Review* 84(4): September.

You can use this paper as a rough guide. However, Card & Krueger conducted their own survey of fast-food restaurants, which is not what I'm asking you to do! Instead, you can use public survey data to investigate the impact of a different minimum wage increase which occurred more recently (in the past 10-15 years) or some other policy. Just as in any difference-in-differences, you will need at least one "treatment" state, and at least one sensibly chosen comparison state. Examples of analyses you could do (see the attachment for a full list of state minimum wage increases in the past decade):

- NJ-PA again: After a series of smaller minimum wage increases, NJ started increasing its minimum wage substantially in 2019. In comparison, PA has not changed its minimum wage, which is set at the federal minimum (now $7.25).
- Illinois began increasing its minimum wage by about $1 per year in 2019. Neighboring Indiana has not.
- Maine has one of the highest minimum wages in the country at $15.10 (as of 2026) more than double New Hampshire's ($7.25). They began to diverge beginning in 2017. Did this change affect employment? Did low-skill workers start moving to Maine from New Hampshire?
- Washington DC has the highest minimum wages in the country ($18.10 as of 2026), and began increasing it in around 2018. What's a good comparison group? Maybe other US cities? But you'd have to check that they don't have their own minimum wage increases.
- Other Policies. **You do not have to study a minimum wage increase at all!** There are many state policies you could study. For example, in the 1990s, some states started requiring that you make adequate progress towards completing high school to receive a driver's license. The impact of such policies on education, for example, could be estimated as a difference-in-differences. One year a group studied the effect of the elimination of "abstinence only" education in 2009 NC on teen pregnancy. That is a bit harder, because it requires getting data on teen pregnancy, though the IPUMS data can at least identify young mothers.

  And some diff-in-diffs do not even involve geography – just a treatment and comparison group.

- In general, what do you need for a difference-in-difference?
    - **A treatment group** – a policy that applies to some but not others. This could be a minimum wage that applies in one state, but not another.
    - **A comparison group** – a region or group to which the policy does not apply, and which you are comparing the treatment group to.
    - **A period before and after the treatment applies**. A key thing you need in order to estimate a diff-in-diff is to know when the policy came into effect.
    - **Data that covers that identifies all of this and contains relevant outcomes.** If you want to do a state diff-in-diff, for example, you need data that identify states before and after the treatment comes into effect + an outcome you want to study.

Also:

- If you do a state diff-in-diff, the comparison regions do not necessarily need to be geographically proximate, but you'll have to make a case for your choice of comparison region in the paper using reason and data (things like a balance test, or common "pre-trends"). You also want to be careful to choose a comparison region that does not itself experience a minimum wage increase in the period you study!
- Which state(s) should you choose? It really doesn't matter much, but it might be interesting to think about choosing some state which would give you some additional motivation to write about. For example, Card & Krueger were looking at largely urban NJ and eastern PA; you could look at rural Washington (which increased its minimum wage) + Idaho (comparison). Might the effects of a minimum wage increase be different in this context? This would be a nice basis for a paper.
- As you have seen, a comparison group also does not have to be another state – it could be a group unaffected or less affected by the policy. (Or you could do both and have a placebo.)
- Don't forget that, through data choices you will make, you will also be able to select
    - What outcomes you will look at. You do not have to look at employment. You could look at something novel like school enrollment or health or fertility. All you need is an argument for how the minimum wage could potentially affect that outcome. All papers might want to look at wages as an outcome as well.
    - The sample of people you will focus on. Teenagers, or other workers with little work experience (you know how to calculate a measure of that!) would be good to look at. But if your sample allows it, you could focus on the impact on particular minority groups (African-Americans, Native-Americans, foreign-born). You could potentially also create an alternative comparison group within state, kind of like Card and Krueger did. Some groups may be too small to feasibly study with the data I am recommending, something you will find out by trial and error.

Data

One data set I am recommending for the analysis is something called the American Community Survey (ACS). This is an annual demographic survey which includes variables like wages, industry & occupation, education, race, ethnicity, sex, and, importantly, state. They also contain some sub-state geography which is a little complicated to use but could be useful in some cases (for example, if you want to create an "urban" comparison area for Washington, DC, or you just want the eastern part of PA as a comparison group for NJ). The data essentially begin in 2006 (there are more limited samples 2001-2005) and are typically available up to about a year prior to the current year, at ipums.org.

I posted a guide to obtaining ACS data from IPUMS. It requires getting an account, but this is not hard to do. Once you have an account, you:

1. Select the years of data you want. I recommend requesting a few years before and after the policy you are studying, so you can look for "pre-trends" and maybe look at some longer-term impacts (if you want to).
2. Select the variables you want. I will have more detailed recommendations but note that you definitely need state identifiers (specifically, I'd recommend state FIPS codes.)
3. Select the sample you want. This part is technically optional -- also be done within STATA instead – but you might want to keep only data from the states you are interested in, which will make your dataset much smaller. (The ACS is fairly large – it is a 1% sample of the US.). This can be done under the "select cases" button on the final screen of the data extraction process (see picture on next page).
4. Request a STATA dataset be created. This is important : otherwise it will send you data in some other non-STATA format. (See picture on next page).

See below for:

- A picture data extraction from the IPUMS website (Video with more details on Canvas)
- Even further below: other data you could potentially use besides ACS data.

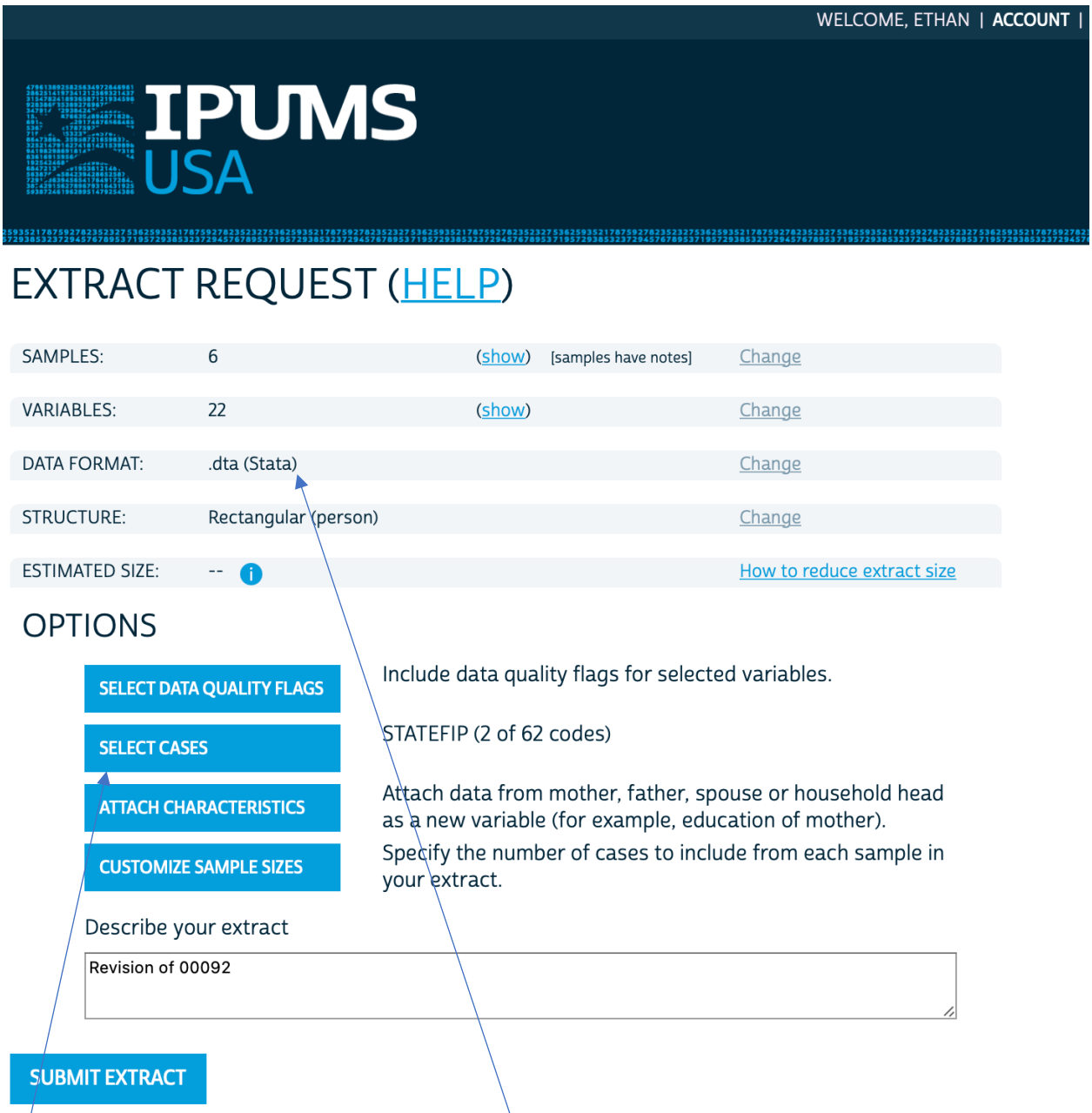


Figure from IPUMS website: Make sure "data format" says "STATA" otherwise click on the "change" button!

"Select cases" allows you to pick out the states (or other sample restrictions) you want ahead of time.

Putting your data together is going to be one of the bigger challenges of this project. You'll figure out through trial and error that certain variables are not useful or don't have what you think that they have; certain samples may be infeasibly small, etc. We will provide assistance with your data construction.

Other data ideas

Current Population Survey. (CPS) This is the data we used in DE1 and DE2. It is very similar to the ACS but (1) it is monthly instead of yearly and (2) it has smaller sample sizes. It is also available in a separate place on the IPUMS website. These are also the data Card & Krueger used in their secondary analysis of teenage employment. These data have some advantages. They have interesting survey modules that ask more detailed questions about particular topics (such as school enrollment, disability, health insurance, and other things). They go back further in time than the ACS. They also theoretically have more accurate information on wages, because there are questions identifying workers who are paid 'by the hour.' However, they survey fewer people each year than the ACS and, though they have a similar extraction procedure to the ACS on the IPUMS website, they are more complicated to use (survey questions change from month to month)

Behavioral Risk Factor Surveillance System (BRFSS). Contains data on substance use and some health information. It is a pretty good option for certain research questions. Unfortunately, it does not have a nice data interface like the IPUMS data, but I can help you use these data if you have a question that these data would help with.

The *National Health Interview Survey* (NHIS) we used in DE3 is available on IPUMS, but it unfortunately does not identify state! So to use it, you'd have to come up with a diff-in-diff that did not involve geography.

Other ideas… find your own data. Many national and local governments post their own data you could use. I can help get it into a useable format for you.

It could be more work to put together alternative data than the ACS. Come to me with a proposal early if you are considering this and develop a simpler plan in case this topic falls through.

Alternative Topics

As I said above, you don't even have to study the impact of a minimum wage increase – you could study some entirely different policy! However, your topic has to fit reasonably well into a difference-in-differences framework, that it needs to have a clear "treatment" and a "treatment group" and a "comparison group." State difference-in-differences framework is one that could make effective use of the datasets I recommended above, so state policies broadly are something you could consider studying. You may have knowledge of interesting state policies from other parts of the country you have been to or lived in. (

Finally, you also don't technically have to use geography; what you need is a "treatment" and "comparison" group.

Additional Notes – programming in STATA

My recommendation is that you have (at least) two STATA programs

1. One which constructs your dataset, e.g., selects or identifies the main sample you want to use, merges in any new variables, constructs variables that will be used in your regressions (ln wage) etc. The last command of this program will be something like "`save final_data.dta, replace`".
2. Another which actually runs the regressions and makes graphs. These could be two separate programs – one for tables and one for graphs – but doesn't need to be. In any case, having a "final" dataset will make it easier to divide up the work of doing the data analysis – multiple group members could work with final_data.dta.

So that I can consult them in case of questions, these log files should be turned in with your completed paper. I will be looking for programs that are well documented (e.g., has comments).

These programs should also be brought with you if you come to office hours to discuss the project. Within the actual paper, you'll want to present your estimated models and also descriptive statistics (mean, standard deviation) for all of the variables used in these models. Note that you want to *look* at these statistics to see if they make sense!

Code library available! On canvas, I have some code up for things that students often want to do, such as adjust wages for inflation.

Appendix A. Variables you should obtain from an ACS extract.

I will post a video on creating ACS extracts from the ipums.org website, but here I will also provide a list of variables you may want to include in your extract, especially if you are looking at minimum wages. Variables marked with a ** are ones that I think basically everyone should get:

Under Household → Geographic

STATEFIP – this is a standardized state code, which will allow you to merge to other data if necessary. It is a key part of the analysis, which is a state policy evaluation! **For most projects this is the only geography you will need. However some projects might benefit from other variables like:

CITY – city identifiers.

MET2013—metropolitan area codes, defined with roughly consistent geography over the years. Metropolitan areas are larger than cities, as they include the attached suburbs.

DENSITY – population density of the area

COUNTYFIP – an identifier for each county, that in combination with STATEFIP is unique and can be used to merge with other data.

…and even more advanced versions:

PUMA – this is the most detailed geography of the ACS. Unfortunately, it varies over time and is not exactly related to any other geography. The IPUMS website has the exact maps of these, but it is very hard to use.
CPUMA is a version where the geography has been held roughly consistent over time.
MIGPUMA1 – where you lived last year

Under Person → Demographic
**AGE – as you are likely going to be looking at workers, you will probably the very old and children at least, and you may want to focus in on young workers.

You may also want other variables here, like SEX, depending on the project, or marital status.

Under Person → Race…
Contains a variety of race/ethnicity variables, some of which you may want for your project, depending on what you're looking at. Basic race categories are in the variable "RACE," whereas Hispanic ethnicity is in "HISPAN."

RACE is a bit complicated by the fact that many people report multiple races. So, if you want single race categories, you could use their collapsed versions, things like "RACBLK" which reports people who report "any" black race. You could also try using their "PRED…" variables, which attempt to collapse race down to single-race categories. (Whatever you choose, describe it in your "data and methods" section.)

CITIZEN identifies immigrants.

Under Person → Education…
**EDUC – educational attainment. Not everyone may want this, but it is good for maybe identifying a group more likely to be impacted by the minimum wage.

**SCHOOL – enrollment in school (or not). To help identify workers, we often drop those who are enrolled in school. Alternatively, school enrollment could be an outcome for teenagers you could examine, or something like that.

There are also other interesting variables here.

Under Person → Work…
**EMPSTAT – is the key variable measuring whether people are working. This might be used to construct the main outcome in most projects.

CLASS – measures whether they are entrepreneurs, among other things

OCC/IND – could be used to identify something like fast food workers. The problem is that these codes change over time. A better approach might be to use something like IND1990, which could be used to pick out restaurant workers. (IND1990 = 641) consistently over time. (The main weakness of picking a sample this way, though, is that the minimum wage

may force people out of the sector, and so they won't be captured by their industry. However, you may be able to test that by asking if the number of workers changes.)

**WKSWORK2 – weeks worked, in categories. I will give you code to turn this into a measure of weeks worked, or else you can just impute it from the category midpoints. The reason you need this is to be able to construct hourly wages
**UHSWORK – usual hours worked per week. Needed for the same reason

Under Person → Work…
**INCWAGE –wage and salary income. Hourly wages can be constructed as INCWAGE/(weeks * UHSWORK) (see above). Hopefully you will be able to show that the min wage increased hourly wages in your sample.

Other variables may be of some interest
**INCBUS00 – business and farm income. This is a technical detail, but I often drop workers with substantial business and farm income – they are probably not workers, but business owners. It could also be used as an outcome.

There are other variables that could be used as outcomes here, like POVERTY, or a variety of government supplemental income program, or FTOTINC.

Under Household->Technical
**CPI99 – multiply wage variables by this to convert to constant 1999 dollars.

Summary: minimum set of variables you probably want in any extract:
STATEFIP, CPI99, AGE, EDUC, SCHOOL, EMPSTAT, WKSWORK2, UHSWORK, INCWAGE, INCBUS00.

Appendix B. State minimum wage database

From https://www.dol.gov/agencies/whd/state/minimum-wage/history .

I have put this into a slightly easier to read excel spreadsheet on Canvas, which was up to date as of Fall 2026.

Part 1:
Project Proposal

Project Proposal Assignment/Getting Started on Your Research Projects

To get started on your projects, I would suggest browsing through the IPUMS data and the database of minimum wages and brainstorming ideas for topics. Despite the assignment title, you don't have to study the impact of minimum wages – you could study another policy – but it should fit into the difference-in-differences framework and use data that you can access (so, probably another state policy that could conceivably affect workers would be OK.)

The proposal is the first thing that is due! On the due date, turn in a sheet with both:

- **A preliminary "abstract"** A short description of your project idea so far. Basically this will say what your specific "Y" variables are, and what policy change you are studying, and what your comparison area(s) is (are), and maybe what controls you are thinking about.
- **A table of means** and standard deviations of your Y variables and maybe any controls you have so far. This is a first draft of the "Table 1" of your paper. (Table 1 should always be the table of means!!!). It could also be the start of your "balance" test, by showing the means separately for the "treatment" and "comparison" states before the policy. This table does not have to be super fancy yet, though; I just want to see that you have made some progress getting an extract into STATA.

You are also not necessarily locked into your final approach yet, I just want to see that you have made some progress thinking about it so I can give you feedback. For example:

> "**Are there Gender differences in the effects of minimum wages on employment? Evidence from a 2019 NJ Law**"
>
> In 2019, New Jersey increased it minimum wage to $11 from $8.60. In this paper, we will ask if there are gender differences in the effect of minimum wages on employment rates, using Pennsylvania (whose minimum wage has remained at the Federal $7.25 since 2010) as a comparison state. The idea behind this is that women are considerably more likely than men to work in restaurants and other low-wage retail jobs, which are known to be heavily impacted by minimum wages. We will focus on workers with a high school education or less, ages 16-39.
>
> In our analysis, we plan to probe the sensitivity of results to controls for demographics and possibly population density.

Means by Gender, State and Year.

	Women		Men	
	NJ	PA	NJ	PA
	A. Before the Minimum Wage Increase			
Wage>=$11	0.4801	0.501	0.6389	0.6272
(in Nominal $)	[.5]	[.5002]	[.4806]	[.4837]
Ln(Wage, 2021$)	2.538	2.5045	2.7469	2.6594
	[.6117]	[.7087]	[.7301]	[.6811]
Employed	0.4988	0.5261	0.5539	0.6035
	[.5002]	[.4994]	[.4972]	[.4892]
Not in Labor	0.4371	0.4034	0.3734	0.3208
Force	[.4962]	[.4907]	[.4838]	[.4668]
Retail/Low Skill	0.2335	0.2727	0.1289	0.1378
Service	[.4232]	[.4454]	[.3352]	[.3448]
Age	25.01	25.061	25.5487	25.6418
	[7.7168]	[7.2824]	[7.5714]	[7.2472]
Years of Education	10.847	11.1402	11.0573	11.1143
	[2.1842]	[1.5809]	[1.967]	[1.9173]
Observations	1556	2696	1958	3584
	B. After the Minimum Wage Increase			
Wage>=$11	0.5349	0.6618	0.6915	0.7509
(in Nominal $)	[.4998]	[.4736]	[.4625]	[.4327]
Ln(Wage, 2021$)	2.5409	2.6441	2.7315	2.8024
	[.6317]	[.6583]	[.7595]	[.7481]
Employed	0.4238	0.5496	0.5544	0.5883
	[.4945]	[.4977]	[.4973]	[.4923]
Not in Labor	0.502	0.3926	0.3642	0.3341
Force	[.5004]	[.4885]	[.4815]	[.4718]
Retail/Low Skill	0.2218	0.3284	0.1561	0.1541
Service	[.4158]	[.4698]	[.3632]	[.3611]
Age	24.6198	25.1889	24.9436	25.438
	[7.6891]	[7.3523]	[7.3131]	[7.0817]
Years of Education	10.7557	11.1081	11.0229	11.1729
	[2.2465]	[1.8907]	[2.0768]	[1.8388]
Observations	697	1325	858	1760

Data Source: American Community Survey 2015-2020 (Ruggles et al., 2021); sample includes men & women with <=12 years of education age 16-39. Calculated Using Person Weights. Standard deviations in brackets

Aside on this table: I made this table in excel after rearranging output of the outsum command. At this point, it is not necessary that you make such a fancy table -- you can give me STATA output. At this point, I just want to make sure everyone has created their own data extract. However, for the final project that is not the case. I repeat: **DO NOT PUT RAW STATA OUTPUT IN YOUR FINAL PAPER!!!**

For more on the "outsum" command: see separate handout or the balance test lab.

Notice why this table is helpful: we can already see some weird things in this sample I have. The share above the minimum wage and the real grows faster in PA than in NJ! This may reflect the noisiness of the wage data in the ACS, or differences in characteristics (which could be controlled for, or just noise (this is only a 10% sample of the ACS – ultimately, I should get the whole sample.)

Some Data Reminders / Tips for the Analysis

- Look at your data and codebook closely before doing any regressions. Sometimes there are strange values that represent missing value codes such as: negative numbers for variables that should only be positive, or implausibly large values. (In the ACS 999999 is the income value for people who have no income!)
    - That is part of the motivation for asking for the table of means. Use, for example the "sum" command to pick out unusual values or missing data.
    - Other commands: "codebook *variable*" works on some data that have labels, tab *variable*" – tabulate the values (only works on variables with a small number of values); and "browse *variables*" to literally look at the data, or specific set of *variables* you ask for.
- It is OK to explore the data but remember that ultimately you should do everything using a program ("do" file). Do not do your final analysis using the "command line." You can also work with my sample data, but eventually get your own extract.

Other Things to watch out for

1. Some categorical variables do not have meaningful units. This include "educ" in the ACS (which can be converted to "yrsed" using a program up on Canvas) and what might be a key one for you, "empst." Do not put these into a regression without first redoing them in some way that makes them have meaningful units.
    a. For example, you might want to convert empst to a dummy for employment.
2. Similarly race, marital status, sex, religion, and region tend to be meaningless variables until converted into dummy variables.

3. On wages, I like ln(wage) rather than wage as an outcome. You will quickly discover that many people reportedly earn below the minimum wage, as you see in this table. So you may want to create a different kind of variable also, like the fraction earning above the minimum wage.
    a. ****IMPORTANT!**** Be careful using inequalities in the definition of dummy variables: STATA unfortunately treats missing values as "a very large number." For example,

    gen wage_over_11 = hrwage>=11

    would create a dummy that is equal to 1 if your wage is over $11***... or if your wage data are missing*!** Instead, the correct syntax would be:

    gen wage_over_11 = hrwage>=11 if hrwage~=.

4. (optional) You can use the "person weights" if you want to. Add a [w=perwt] to most commands and it will weight them up to correct for any non-randomness in the sample. One caution: the person weights in 2020 are considered dubious. So consider not using the person weights at all if you are using 2020 data. Whatever you choose to do, just say it in your (future) tables and data and methods section of your paper.

Part 2:
Balance Test/Data and Methods Section

Data and Methods Section
Balance Test

Every paper should have a section which talks about the data construction and estimation approach. This should be written for the so-called "well-informed undergraduate" (something you can relate to!) who is unfamiliar with your project. Essentially, the goal would be to explain to someone who has taken undergraduate econometrics what you have estimated and how you constructed the data. This means, for example, you should not assume (even though the I the professor, already know) what data you are using. **You don't need to necessarily go over every single detail of the project, but write enough to give the reader a good start on being able to replicate exactly what you did.**

When you talk about the data, it is a good place to mention your table of means. When you talk about methods, it is also good to talk about the assumptions necessary for causal inference, which leans naturally to looking at a balance test. Below are some more detailed thoughts on each.

Describe your **data** and **sample construction**

- Say "we used the American Community Survey" obtained from ipums.org (with a full citation) even though it may seem silly to do so in this case. I know what data you are using in most cases, but you are pretending the reader does not. Also, **it is NOT redundant to mention your dataset by name again in this section even though you have already done so in the abstract and introduction!!** So do it again.
    - Mention what years of data you are using
    - Describe any sample cuts :
    - Altogether, you might be able to say this in a single sentence, something along the lines of "We use American Community Survey (ACS) data (Ruggles et al., 2019) from 2012-2018 for the states of New Jersey and Pennsylvania."
- Describe any additional sample construction in this section ("limited to high school dropouts" or whatever), as well as the construction of key variables. Here you'll have to use your judgment about what is "key." Your outcomes are all "key," but obscure controls may not be.
- Make sure to describe any variables you merged into the data.
    - Related to this: **NEVER REFER TO STATA VARIABLE NAMES WITHOUT AT LEAST DEFINING THEM FIRST.**
        - BAD: "We used *empst* to construct our main left-hand side variable. In particular, we examine a dummy for *empst==1*."
        - BETTER: "We used *empst*, a categorical variable which measures the employment status, to construct our main left-hand side variable. In particular, we examine a dummy for *empst=1*, for employed"
        - Or, in fact, you could just say "Our main right left-side variable is a dummy for being employed, constructed from the employment status

variable in the main data." I would only use STATA variable names to save space if it is going to be repeated many times in the paper.

- Mention means. Once you introduce the data and sample, you can make passing reference to the table of means, as in "Means and standard deviations of the main variables used in the analysis are shown in Table 1." You don't need to say much more than that unless there is something important to point out.

Describe your **methods**

- This could be part of one joint "data and methods" section.
- **Include at least one estimation equation**. That is, a stylized "econometric model" as in "We estimate

$$\ln Wage_i = b_0 + b_1 Post2013_i + b_2 NJ_i + b_3 NJ \times Post2013_i + u_i$$

where Post2013 is dummy for observations after NJ's 2013 minimum wage increase (zero before), and NJ is a dummy for NJ (zero for observations in PA)"

…for example.

  - **Looking ahead: the estimation equation can instead appear elsewhere in the paper, but do not include equations in the introduction.**

- **Say in plain English words (that is NOT just using "u"s or the word "error") the assumptions needed for causal inference**. This might be a good way to setup your balance test (see below), which assesses the validity of these assumptions to the extent that you can observe. For most of you this is going to be some kind of "parallel trends" assumption, as in

  "We assume that absent NJ's minimum wage increase, ln(wages) would have increased by the same amount in NJ and PA.

- Describe how you calculate the standard errors here, and if you are using estimation methods beyond OLS. Most of you will likely estimate using standard errors robust to heteroskedasticity.

Balance Test

The description of the assumptions for causal inference leads naturally to a balance test which challenges those assumptions. One approach you could take to the balance test is to simply run a series of regressions that replace your main outcome variable with

demographic or other controls that you are considering in your analysis. (An even simpler approach is shown below…). To borrow from the estimation equation above for the NJ/PA case, essentially you could estimate:

$Control_i = b_0 + b_1 Post2013_i + b_2 NJ_i + b_3 NJ \times Post2013_i + u_i$

Where "control" is from a list of demographic or other controls. Basically, you want to know if any of the "b_3"s from such regressions are statistical significant and sizeable. This is very similar to what you did in DE3, except that you now are running a "difference-in-differences" instead of a simpler treatment-comparison difference (but see below for something even similar to DE3 you could do.). So you're regressions will look something like (assuming the variable names in the equation match the STATA variable names):

```
local append "replace"
foreach var in list_of_controls {
  reg `var' post2013 nj nj_post2013, robust
  outreg2 nj_post2013 using balance_test, `append' excel noaster
  local append "append"
}
```

Where "list of controls" is actually a list of controls, and "balance_test" is a file name that could be anything you want.

Examples. This type of balance test for a difference-in-differences in modelled by a paper by Susan Dynarski (2003) on the impact on schooling of the end of social security survivor benefits for children aged 18-22. Essentially, Social Security used to pay for kids' college if their father died before a kid turned 18, but this stopped in 1982. The "diff-in-diff" uses before / after 1982, and uses kids whose father didn't die as a comparison group for those whose father did die. Her "Table 1," on the next page, shows the difference-in-difference for a bunch of controls is not significant. I will post her programs on the course page.

While this is definitely not required, a nice feature of Dynarski's table is that it is merged with the table of means. Essentially, she shows the means for the four groups (before/after x treated/untreated) and then adds a fifth column with the b_3 regression coefficient from the diff-in-diff, along with the standard error. The first four columns can be created with an "outsum" command, while the fifth she would have to cut and paste from the "outreg2" command into the same excel worksheet.

TABLE 1—NLSY SUMMARY STATISTICS

	High-school seniors 1979–1981		High-school seniors 1982–1983		
	Father not deceased	Father deceased	Father not deceased	Father deceased	Difference-in-differences
Household income ($2,000)	54,357	32,875	50,842	32,298	−2,938
	(537)	(1,839)	(788)	(2,958)	(4,816)
AFQT percentile	60.50	58.18	52.87	44.90	5.65
	(0.51)	(2.36)	(0.91)	(3.92)	(5.33)
Black	0.135	0.235	0.151	0.297	−0.046
	(0.007)	(0.036)	(0.011)	(0.063)	(0.068)
Hispanic	0.051	0.055	0.062	0.059	0.007
	(0.004)	(0.020)	(0.007)	(0.032)	(0.026)
Father attended college	0.331	0.184	0.299	0.158	−0.006
	(0.009)	(0.033)	(0.014)	(0.050)	(0.079)
Mother attended college	0.238	0.127	0.203	0.166	−0.074
	(0.008)	(0.029)	(0.012)	(0.050)	(0.085)
Single-parent household	0.153	0.787	0.194	0.837	−0.009
	(0.007)	(0.035)	(0.012)	(0.051)	(0.071)
Family size	4.77	4.40	4.71	4.34	0.00
	(0.03)	(0.18)	(0.05)	(0.27)	(0.31)
Age in 1988	25.95	25.92	23.95	23.95	−0.03
	(0.02)	(0.09)	(0.02)	(0.09)	(0.14)
Female	0.483	0.488	0.471	0.485	−0.009
	(0.010)	(0.043)	(0.015)	(0.069)	(0.097)
Attend college by 23	0.502	0.560	0.476	0.352	0.182
	(0.010)	(0.043)	(0.015)	(0.066)	(0.096)
Complete any college by 23	0.487	0.560	0.459	0.361	0.171
	(0.010)	(0.043)	(0.015)	(0.066)	(0.097)
Years of schooling at 23	13.41	13.44	13.25	12.90	0.380
	(0.03)	(0.13)	(0.05)	(0.20)	(0.296)
Number of observations	2,745	137	1,050	54	3,986

Notes: Means are of NLSY poverty and random samples, weighted by 1988 sample weights. Income and household composition measured during senior year of high school. AFQT is age adjusted; see Data Appendix. Standard errors are in parentheses. Standard errors in the difference-in-differences column adjusted for clustering at the household level.

Balance test from Dynarski (2003), combined with table of means.

Another approach: look at differences beforehand.
Card & Krueger, you may recall, just look at the differences in "wave 1" (before the minimum wage increase). They did this because they knew they had a balanced panel so average (predetermined) firm characteristics were not changing over time by definition. The regression setup for this would be similar to the above, except to just look at the "NJ" (or treatment area) dummy coefficient beforehand, as in:

```
local append "replace"
foreach var in list_of_controls {
  reg `var' nj if post2013==0, robust
  outreg2 nj using balance_test, `append' excel noaster
  local append "append"
}
```

Again, it would be possible to combine this with the table of means, kind of like Card & Krueger did:

Table 1. Means and Pre-Minimum Wage Increase Balance

	Women		
	NJ	PA	Difference
	(1)	(2)	(3)
Before the Minimum Wage Increase			
Wage>=$11	0.4801	0.501	0.0124
(in Nominal $)	[.5]	[.5002]	(0.0159)
Ln(Wage, 2021$)	2.538	2.5045	0.0808
	[.6117]	[.7087]	(0.0232)
Employed	0.4988	0.5261	-0.0508
	[.5002]	[.4994]	(0.0105)
Not in Labor	0.4371	0.4034	0.0548
Force	[.4962]	[.4907]	(0.0104)
Retail/Low Skill	0.2335	0.2727	-0.0274
Service	[.4232]	[.4454]	(0.00815)
Age	25.01	25.061	-0.203
	[7.7168]	[7.2824]	(0.158)
Years of Education	10.847	11.1402	-0.230
	[2.1842]	[1.5809]	(0.0422)

Data Source: American Community Survey 2015-2020 (Ruggles et al., 2021); sample includes women with <=12 years of education age 16-39. Calculated Using Person Weights. Standard deviations in brackets (columns 1,2) , standard errors in parentheses (column 3).

This shows a bunch of pre-minimum wage increase differences – for example years of education – that you could think about controlling for in your main regression (in a way that I will explain later.). You could also have another panel for after the minimum wage increase, as Card & Krueger did, but that would not be for balance, just to show the means for those data as well.

Even if you do your test this way, remember to say that it is not strictly required that these pre-treatment means be the same for the treatment and comparison area in order for the comparison area to be a valid proxy for the counterfactual. The assumption is "parallel trends" NOT "parallel levels." That is why the first, more complicated version of the balance test is potentially more pertinent. Still, the more similar the two states are beforehand, the less likely their trends might diverge for other reasons, which is why there

is some value in looking at this. Alternatively, controls for imbalanced covariates may still help reduce bias.

For example, based on the table above, we might be concerned that if trends in employment rates differ by education, it could bias the estimates due to the pre-existing difference in education. A control for education might help with this.

How these tables were made

These tables were made with the help of the "outsum" (means) and "outreg2" (regression output) commands, described below, and then further edited in Microsoft excel. Use these commands to construct your final tables.

Creating final versions of your tables: outreg2 and outsum

The final version of your paper should not have STATA output, it should have nicely formatted tables (e.g., in excel) that look like a real published paper! No, you do not have to do a bunch of cutting and pasting to get this look: instead, in STATA there are commands that allow you to construct regression tables (outreg2) and tables of means (outsum) directly.

Outreg2

Outreg2 is an installable STATA command which allows you to make regression tables like you see in papers. The first step is to install the command. With a computer connected to the internet, type on the STATA command line (you only ever have to do this once):

```
ssc install outreg2
```

After that, you can use outreg2 **after running any regression**. So first you run a regression, then you type the syntax:

```
outreg2 [list of variables] using [filename], excel [replace/append] [other options]
```

…where (brackets not actually part of the command, by the way):

- [list of variables] - is a list of any subset of variables that were used in the regression. These coefficients on these variables will be shown. By default, all of the independent variables in your regression are shown. But if you, say, have a lot of dummy variables you might want to suppress these coefficients.
    - Note that in some versions of outreg2 listing the variables which you want to show does not work. So instead to hide coefficients you must use the "drop" option, described below.
- [filename] - can be any name you want to give it -- although if it has a space in the name it should be in quotes. The regression table will appear in this file.
- `excel` -- output the table as an excel file
    - By the way, you don't have to output it as an excel file. The default is a text file, which you can also read into excel. Open the file from excel directly (Open… and choose the "All Files" option), and read in a "delimited" file.
    - There is also a **`latex`** option for the hardcore. Like the excel option, it does not create a final version of the table, just some latex starter code that gives the outlines of a possible table.
- [replace/append] -- To create the first column of the table use "replace." To add columns to the table, type "append." Put differently, the first time you use outreg2

when you are creating a given table you type "replace." Every time thereafter, you type "append" instead.

- Other options include:
  - **`ctitle(`*`text`*`)`** - creates a column header containing the text of *title*. This is a good way to keep track of what was in each regression. For example you might have column headers "OLS" and "Fixed effects." You can continue a column header to a second line by have two sets of text separated by a comma, as in, `ctitle(OLS,Females Only)`
  - **`noaster`** -- removes the "*"s that indicate statistical significance levels.
  - **`drop(`*`list of vars`*`)`** – do not show the coefficients on the specified *list of vars*. This is especially with long lists of controls – such as fixed effects – where we are not really interested in the coefficients, just the knowledge that that list of variables is controlled for. It is especially useful in combination with the "addtext" option.
  - **`keep(`*`list of vars`*`)`** – similar to previous.
  - **`addtext`** – allows you do add that bottom row with things like "Fixed effects? Yes." The syntax is simple: `addtext(`*`description1`*`,`*`value1`*`, `*`desciption2`*`,`*`value2`*`,…)`. The "descriptions" are the things that go in the leftmost column (like "Fixed effects?), while the "values" are the things that vary from column to column ("Yes" "No" etc.). Also:
    1. Notice that you could have several of these, for example, "`addtext(Year Effects, Yes, Fixed Effects, No)`"
    2. Notice also that the value of this command is being able to repeat it across *different* regressions. So if the above was used after a regression without fixed effects, then you might run the regression again with fixed effects, but instead have "`addtext(Year Effects, Yes, Fixed Effects, Yes)`"
  - **`addstat`** -- if you want to add an F-test to the table, this is how you'd do it:
    1. After your regression, do your "test" command.
    2. Then do your "outreg2" command, and after the comma have "`addstat(`*`description1`*`,r(F),`*`description2`*`,r(p))`" Note that "desciption1" and "description 2" are just labels that will appear in the leftmost column of your table that you want to give to the F-stat (which is stored in the variable r(F)) and the p-value (r(p)). So, generically it might be "F-stat" and "p-value." It adds it to the bottom of the table. We can talk in more detail about how to do that if you want. You could also add the root MSE if you wanted, with "`addstat(RMSE, e(rmse))`".

- **`addnote(`*`text`*`)`** -- add table footnotes.
- Known "bugs":
    1. I have noticed that on some machines the "addstat" command only works properly with you put the statistics inside special single quotes, as in "`addstat(`*`description1`*`,` `` `r(F)' ``,*`description2`*`,` `` `r(p)' ``)`". These special single quotes are left quote ( ` -- the key below the "~" in the upper left-hand side of the keyboard) and right quote (' – regular apostrophe). On other computers adding this will crash the program.
    2. The command outputs a special excel "executable" format of excel file, which can theoretically open directly from STATA. However, this differs on Macs vs. PCs:
        - On PCs, you can just click on the output window to pull up the excel table.
        - On macs the "excel" format seems not to work by default as an executable. Instead, you have to open the file directly from the excel File menu ("Open…") after which you can save it in regular excel format.

More generally, typing "help outreg2" on the command line will give you a complete list of options. Example on blackboard.

Finally, you are not done once you have created an outreg table. You'll need edit the output of the outreg2 table even further to give it nice titles, formatting, etc. (I would then save it under a new file name so you don't accidently overwrite it if you run your STATA program again.) <u>The final version of the tables should be self-contained and readable by someone not very familiar with the details of what's in the paper.</u>

Outsum

There is a similar STATA command for making fancy tables of means like the one above: outsum. Like with outreg2, you must first install the command, which does not come standard in STATA packages:

```
ssc install outsum
```

The outsum command creates a table with means and standard deviations (in parentheses underneath means). For an examples, see data exercise 4 – the optional part at the end created the table of means by whether or not you had health insurance.

The syntax for using this is shown on the next page. It is similar to outreg2 except that:

- You don't need to run a regression!
- There is, unfortunately, no "excel" option. Instead, it creates a text file which you can read into excel (by opening "all files").
    - I think the best option is to use the "comma" option and use a file name with a ".csv" at the end, which make the file directly openable in excel. For example:

```
use alcohol.dta, clear
gen emp = empst == 1
gen female = sex==2
outsum emp days momhgc dadhgc using means.csv, comma replace bracket ///
ctitle("Overall") nol

outsum emp days momhgc dadhc if fem==1 using means.csv, comma append ///
bracket ctitle("Women") nol

outsum emp days momhgc dadhgc if fem==0 using means.csv, comma append bracket
ctitle("Men") nol
```

This creates a table of means with columns for overall, women, and men in a file called "means.csv" which should be able to open in excel, and should look something like the table of means above in this handout. The optional "`bracket`" option uses square, rather than round parenthesis for the standard errors, as was in the table above. You could follow up by putting in nice title formatting, etc. (and then actually saving it in excel format.) "`nol`" is an option which hides the long variable labels; I have found the tables format better if you

put in this option. Of course, to actually make the table look like it does above, you would have to edit means.csv and give it nice labels and save it as an excel file.

I can provide additional help on this if you need it, but see if you can't figure it out yourself (use "help outsum.")

The detailed syntax is:

```
outsum [list of variables] if sample==1 using filename.csv, replace ///
       nol bracket comma ctitle("column header")
```

[list of variables] is not actually a thing you type in, you replace it with the list of variables you want to take the means of. Similarly, filename.txt is any filename you want to give it. The other options just make the output easier to deal with in excel

replace -- use this to create the table of means. If you want to add columns to it, change it instead to "append." See example below.

nol – optional – it says show the variable names rather than the variable labels.

bracket – puts the standard deviations in brackets "[]" rather than parentheses. Good for distinguishing for standard errors.

comma – saves it as a comma delimited file. You can open such a file in excel and "save as…" an excel file.

ctitle – Put custom "column header" at the top of the column.

Part 3:
Results

Results and Graphs
Collapse and Event Study Pictures

For the next step of your project, it would be useful to try to write up (at least a preliminary version of) your results. If that proves infeasible at this point, at least put together the main table and graph that are the most central to your analysis.

Graphs

- All papers should include at least one graph that is relevant to the estimates. In most cases, this will likely be an "event study" type of graph, showing how the outcome changed in the treatment and comparison areas (groups) both before and after the treatment.
    - This is easier if you first "**collapse**" (a STATA command) the data down to the means by year and treatment/comparison group. **See below** for example.
- It often makes sense to introduce at least some graphs before tables. Graphical evidence is simpler and more transparent and tends to be easier to introduce. In your case, your likely graph will ideally show a lack of "pre-trends" which it is useful to show before turning to the main results.
- Like tables, graphs should be readable. Graph options like title("…") and note("…") and similar xtitle and ytitle options should be used. Again, use the STATA help menus.

Table Tips

- All papers should include at least three tables: descriptive statistics (means and standard deviations), balance test (can be merged with the table of means), and main results. Some tables may include more (e.g., robustness checks).
- Never include raw STATA output in your paper
    - Instead, use "outreg2" which constructs professional-style tables. More on this in other handouts.
- Tables should be self-contained
    - That is, there should be sufficient notes and or headers to figure out what is in the regression without having to read the paper. Even with the "outreg2" command output, you may need to further edit your excel spreadsheet to make the final version of the table clear. **I recommend "save as…" and changing the file type to an "Excel Workbook" if you do this, otherwise you may accidentally overwrite your edits when you re-run your STATA program.**
- Start simple then add complexity. If you are including specifications with a bunch of controls, first show specification without controls.

- Or, to take more unlikely examples, if you are estimating by IV or logit, first estimate by OLS
- Also, try to keep the sample as consistent as possible across specification, or else provide a "bridge" between specifications that shows the impact of changing the sample.

Writing Tips

- Write in natural paragraphs. (Unlike in this handout!) bullet points and checklist style of writing should be kept to a minimum.
- Stay focused on your core results. You do not need to discuss the coefficient on every control, etc. Instead, all the writeup should be focused on contributing to the main narrative of the paper.
- **All** tables that you use should be referred to by their table number somewhere in the text, and **tables should appear in the order they are referred to in the text.**
- Interpret the magnitude, not just the statistical significance. Statistical significance is not a goal: your project is interesting regardless of what you find.
  - If it is insignificant, and you want to say something further, it is also good to describe the confidence interval.
- Avoid unsupported assertions!
  - If you are a little unsure, ask yourself: do I need to say this? **It is better to say nothing than to say something wrong.**

STATA "collapse" command

The STATA "collapse" command allows you to reduce your data down to statistics (means, standard deviations etc.) by "cells" that you specify. This is useful in this case because you have microdata – the ACS in most cases – but you really are interested in how the averages change by year and state, the level of the minimum wage change. Below is a simple example of a program looking at the 2014 minimum wage increase in NJ compared to PA. I start with the data construction for your reference but below you'll see the "collapse" command

```
/* data construction for reference */
* keep NJ and PA
keep if statefip==34|statefip==42
gen nj = statefip==34 // dummy for NJ

* keep a few years before & after the policy
keep if year>=2011
keep if year<=2017

* only less educated
keep if yrsed<=15

* only less than 40, older than 16, men
keep if age>=16 & age<40
keep if sex==1 // men
```

```
* ln(real hourly wage), 2021 dollars
gen lnrwage = ln(hrwage/deflator)

* employment dummies
gen employed = empstat==1
```

Now, the collapse command:

```
collapse (mean) lnrwage employed [w=perwt], by(nj year)
```

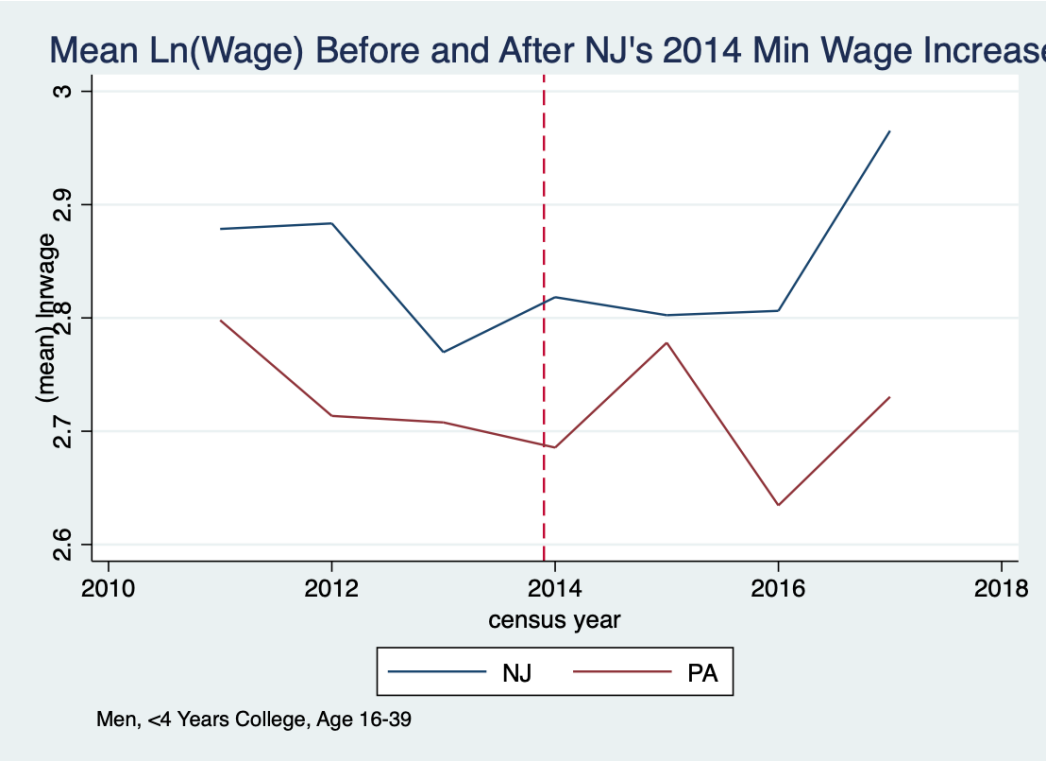
This command "collapses" the data down to just the mean ln(wage) and employment rate by NJ (1/0=PA) and year. The collapsed data have just 14 observations – 7 years each for NJ and PA – and look like this:

nj	year	lnrwage	employed
1	2011	2.87855	.5255272
1	2012	2.88345	.5931177
1	2013	2.769717	.6094129
1	2014	2.818321	.5929292
1	2015	2.802434	.6102242
1	2016	2.806291	.6510258
1	2017	2.965224	.5708674
0	2011	2.79798	.5704111
0	2012	2.713549	.5584464
0	2013	2.707762	.6081174
0	2014	2.685568	.6067631
0	2015	2.778139	.6400695
0	2016	2.634558	.611331
0	2017	2.730391	.6396411

That's it! All the individual observations have been destroyed in favor of these means (below I will describe how to preserve the underlying microdata). The value of this is that we can use this to make a nice graph, to visualize pre- and post-trends in outcomes.

Recall above that I suggested showing the graph first before turning to the tables. Can see why? In one easy to read graph we see both little sign of "pre-trends" beforehand, supporting the comparison group, and then some divergence afterward, supporting an impact of the minimum wage increase, a treatment effect. Thus, all in this simple graph you can understand better what the tables introduced later reveal.

The graph is pictured below:



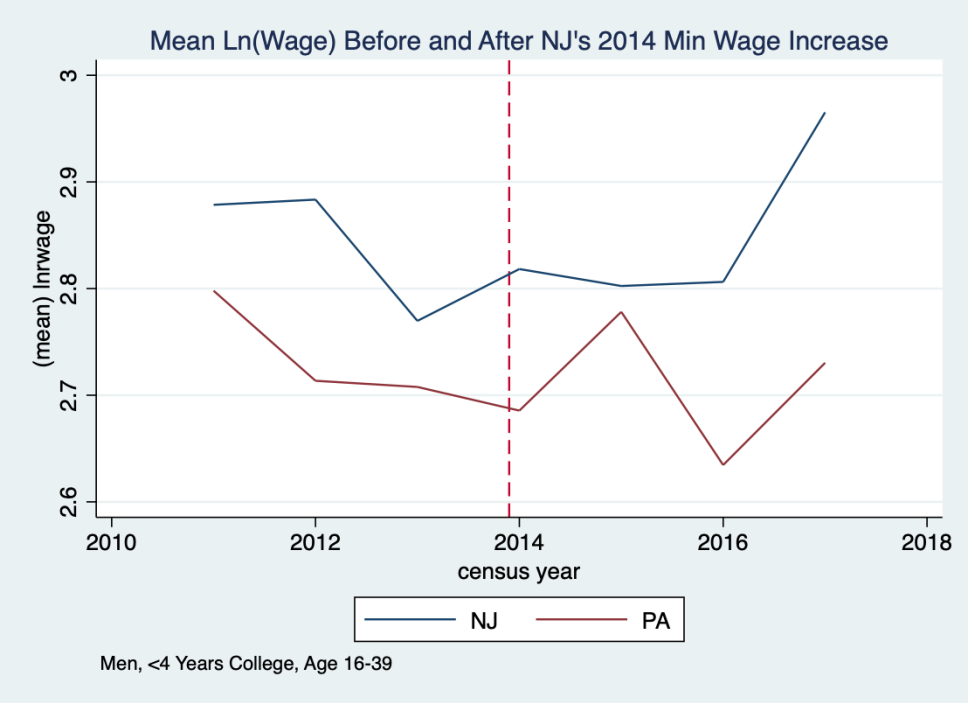

```
gr two (line lnrwage year if nj==1) ///
    (line lnrwage year if nj==0, xline(2013.9, lp(dash))), ///
    legend(label(1 "NJ") label(2 "PA") ) ///
  title("Mean Ln(Wage) Before and After NJ's 2014 Min Wage Increase") ///
            note("Men, <4 Years College, Age 16-39")
```

Notice the details of the syntax:

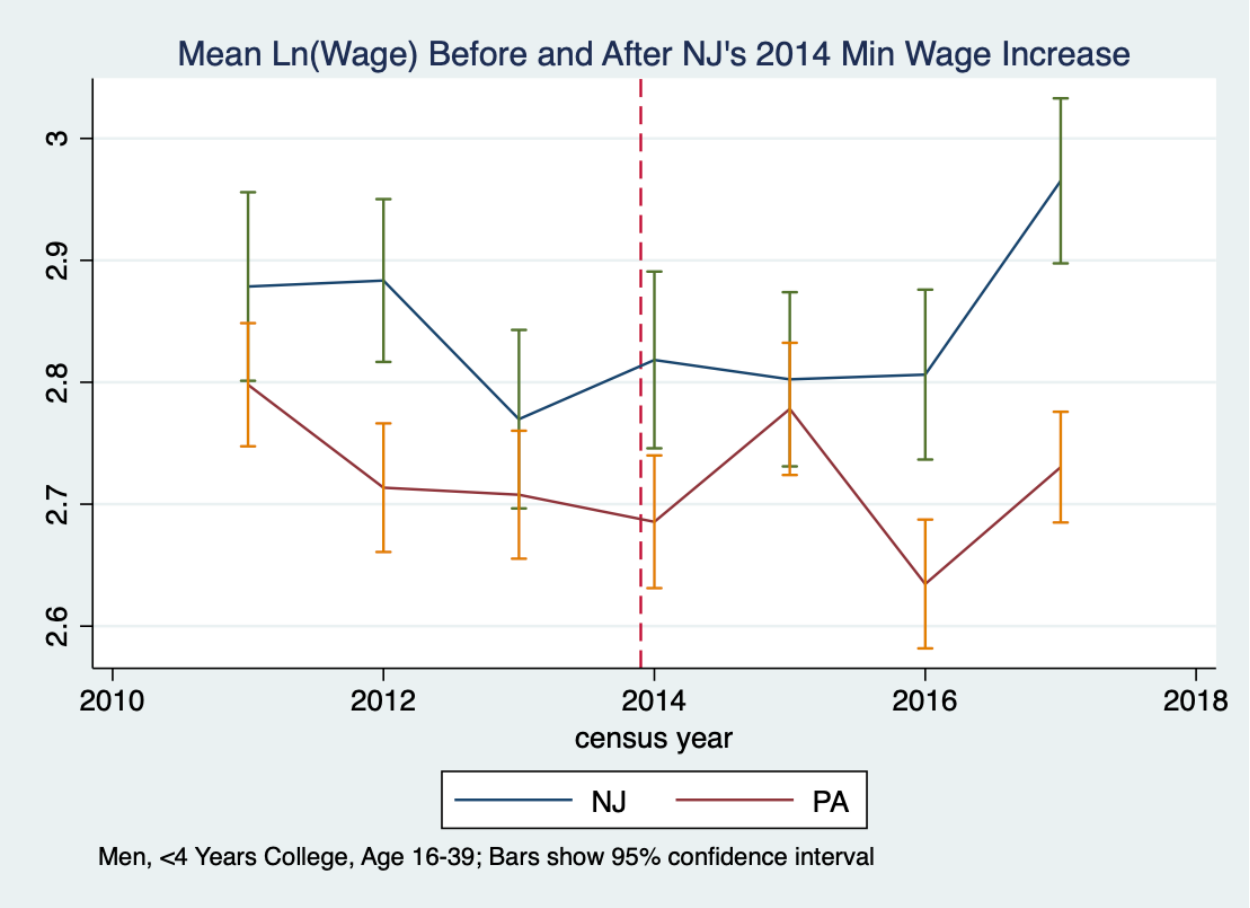
- "gr two" allows you to combine several graphs, which are listed in parentheses.
    - The first graph is `(line lnrwage year if nj==1)`, that is, a line graph between ln(wages) and year for NJ.
    - The second graph is a line graph instead for PA. It also includes the option ", `xline(2013.9, lp(dash))`" which draws a dashed vertical line just before the minimum wage goes into effect in 2014 (2013.9).
- The legend option allows you to label the two lines. The first line is NJ and second is PA. See the syntax above
- I also added a title and a note
- The " ///" lets you split a long command across lines
- The title is a little long; I could make it fit better by shrinking the font size, or by splitting the title over multiple lines instead:

Shrink font size of title, add "size" option:

```
gr two (line lnrwage year if nj==1) ///
    (line lnrwage year if nj==0, xline(2013.9, lp(dash))), ///
legend(label(1 "NJ") label(2 "PA") ) note("Men, <4 Years College, Age 16-39")///
title("Mean Ln(Wage) Before and After NJ's 2014 Min Wage Increase", size(medium))
```



The graph above is sufficient for your analysis. An even fancier thing you could do, which is a little bit involved, is to create confidence interval error bars on the graphs' points with the "rcap" graphing command, which can give you some visual information on whether the pre-trends are significant. The results look like this:



You have to construct these confidence intervals by hand when you do the collapse command. You add to the collapse command statistics for the <u>standard errors</u>:

```
collapse (mean) lnrwage employed ///
         (semean) se_lnrwage=lnrwage se_employed = employed ///
     [w=perwt], by(nj year)
```

Notice the general syntax of collapse:

- *(statistic) name of variable in collapsed data = name of variable in original data*
- So in this case we have added (semean) which calculates the standard error on the mean, and we have created a new variable se_lnrwage, which is the standard error on the mean of the ln(wage). Notice that we cannot call this lnrwage, because lnrwage is already the name of the mean wage in the collapsed data.
- Similarly for the employed variable.

We can then calculate variables that capture the confidence interval as the mean +/- 1.96 times the standard error:

```
gen upper= lnrwage+1.96*se_lnrwage
gen lower= lnrwage-1.96*se_lnrwage
```

Which leads to our graphing command that also plots these upper and lower confidence bars:

```
gr two (line lnrwage year if nj==1) ///
       (line lnrwage year if nj==0, xline(2013.9, lp(dash))) ///
         (rcap upper lower year if nj==1) ///
         (rcap upper lower year if nj==0), ///
     legend(label(1 "NJ") label(2 "PA") order(1 2)) ///
title("Mean Ln(Wage) Before and After NJ's 2014 Min Wage Increase", size(medium)) ///
note("Men, <4 Years College, Age 16-39.  Bars show 95% confidence interval")
```

See the syntax added:

- Rcap upper lower year – the "upper" and "lower" are the upper and lower ends of the confidence interval we constructed
- I added an "order(1 2)" to the legend option, which says only put the first two plots in the legend.

Bonus thought #1: preserve/restore to keep the microdata available.
One problem with the collapse command is that it overwrites your dataset with these cell averages. If you want to make your graphs, and then go back to your original microdata, you can "preserve" your data before the collapse command and then "restore" after you go back to working with the microdata. So to take a previous example, we could have a full set of code that looked like this:

```
preserve
collapse (mean) lnrwage employed ///
         (semean) se_lnrwage=lnrwage se_employed = employed ///
     [w=perwt], by(nj year)
gen upper= lnrwage+1.96*se_lnrwage
gen lower= lnrwage-1.96*se_lnrwage
gr two (line lnrwage year if nj==1) ///
       (line lnrwage year if nj==0, xline(2013.9, lp(dash))) ///
         (rcap upper lower year if nj==1) ///
         (rcap upper lower year if nj==0), ///
     legend(label(1 "NJ") label(2 "PA") order(1 2)) ///
title("Mean Ln(Wage) Before and After NJ's 2014 Min Wage Increase", size(medium)) ///
note("Men, <4 Years College, Age 16-39.  Bars show 95% confidence interval")
restore
```

If we did that, we could then go back to running regressions or whatever, with the microdata, after the restore command.

Bonus Thought #2: Fully differenced event study.
The following is very advanced, and definitely NOT required for your analysis.

Ultimately what we want to know is if there is a statistically significant difference between the treatment state and control state after the treatment compared to before. We want to

know whether any difference was already changing significantly before the treatment (pretrends). We can test this with a very general regression model where the treatment state is interacted with a separate dummy for each year except the year before the treatment. This effectively estimates a diff-in-diff separate for each year compared to the year before the treatment, including in the years before the treatment happened.

For example, suppose you have data on NJ and PA for the 2014 minimum wage increase. The year before the 2014 treatment is 2013, which will be the excluded year. Fully written out, then, we could estimate the following model:

$$Y = \beta_0 + \beta_1 D2011 + \beta_2 D2012 + \beta_3 D2014 + \beta_4 D2015 + \beta_5 D2016 + \beta_6 D2016 + \beta_7 NJ + \beta_8 NJ \times D2011 + \beta_9 NJ \times D2012 + \beta_{10} NJ \times D2014 + \beta_{11} NJ \times D2015 + \beta_{12} NJ \times D2016 + \beta_{13} NJ \times D2017 + u$$

Where DYYYY is a dummy for year YYYY (relative to 2013, the excluded year). Notice that we have all those year dummies plus interactions with NJ. Thus, each $NJ \times DYYYY$ interaction asks how much the outcome changed in NJ relative to PA between 2013 and YYYY – it's a year-specific diff-in-diff. Ideally, all the ones before 2013 are not significant (= "no significant pretrends).

To do this in STATA, we would need to create dummies for all of the years, and then dummies for the interaction of all the years with a NJ dummy. I would probably do this with a loop:[1]

```
foreach y in 2011 2012 2014 2015 2016 2017 {
  gen yrdum`y' = (year==`y')
  gen njXyrdum`y' = nj*yrdum`y'
}
```

You could then run a regression with all these dummies and interactions:

```
reg lnrwage yrdum* nj njXyrdum* [w=perwt], robust
```

```
Linear regression                               Number of obs     =      7,640
                                                F(13, 7626)       =       4.90
                                                Prob > F          =     0.0000
                                                R-squared         =     0.0127
                                                Root MSE          =     .70873

------------------------------------------------------------------------------
             |               Robust
     lnrwage | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
   yrdum2011 |   .0902186    .050365     1.79   0.073    -.0085106    .1889478
   yrdum2012 |   .0057868   .0507454     0.11   0.909    -.0936881    .1052618
   yrdum2014 |  -.0221937    .052836    -0.42   0.674    -.1257669    .0813794
   yrdum2015 |   .0703775   .0528706     1.33   0.183    -.0332634    .1740184
   yrdum2016 |  -.0732036   .0532939    -1.37   0.170    -.1776744    .0312672
   yrdum2017 |   .0226293    .044587     0.51   0.612    -.0647736    .1100321
```

[1] Important! This is all done in the microdata. You are NOT collapsing the data before you do all of this.

```
          nj |    .061955   .0558686     1.11   0.267    -.0475629    .1714729
njXyrdum2011 |   .0186143   .0822418     0.23   0.821    -.1426023    .1798309
njXyrdum2012 |   .1079462   .0783724     1.38   0.168    -.0456853    .2615776
njXyrdum2014 |   .0707984    .083313     0.85   0.395     -.092518    .2341149
njXyrdum2015 |    -.03766    .080822    -0.47   0.641    -.1960934    .1207733
njXyrdum2016 |   .1097777   .0800512     1.37   0.170    -.0471446       .2667
njXyrdum2017 |   .1728781   .0738476     2.34   0.019     .0281164    .3176398
       _cons |   2.707762   .0344761    78.54   0.000     2.640179    2.775344
------------------------------------------------------------------------------
```

The interactions before 2014 are not statistically significant, indicating a lack of significant pre-trends.

Triple fancy! Definitely not required!!! You can plot these coefficients and standard errors. Oh my gosh, the required syntax to do so is painful, though. What you would need to do is: (0) run the regression above (1) create empty variables that are all zeros for the slope and all missings for standard error, and then (2) copy the slopes and standard errors of each year into these variables year-by-year and then (3) collapse these to the year level, and (4) construct the confidence interval and then (5) plot them, similar to above.

To preview the syntax, step (2) uses the fact that for any *variable* in your regression, the value of the slope is stored inside the macro variable _b[*variable*] and the standard error is stored in _se[*variable*]. Now here's the complete STATA syntax:

```
* step (0): run the fully interacted regression above.  To repeat it here:
reg lnrwage yrdum* nj njXyrdum* [w=perwt], robust

* step (1): create empty variables for the slope and standard error
gen slope = 0
gen se = .

*  Step (2): copy the slopes and standard errors of each year into these
variables year-by-year and then
foreach y in 2011 2012 2014 2015 2016 2017 {
  replace slope = _b[njXyrdum`y'] if year==`y'
  replace se = _se[njXyrdum`y'] if year==`y'
}

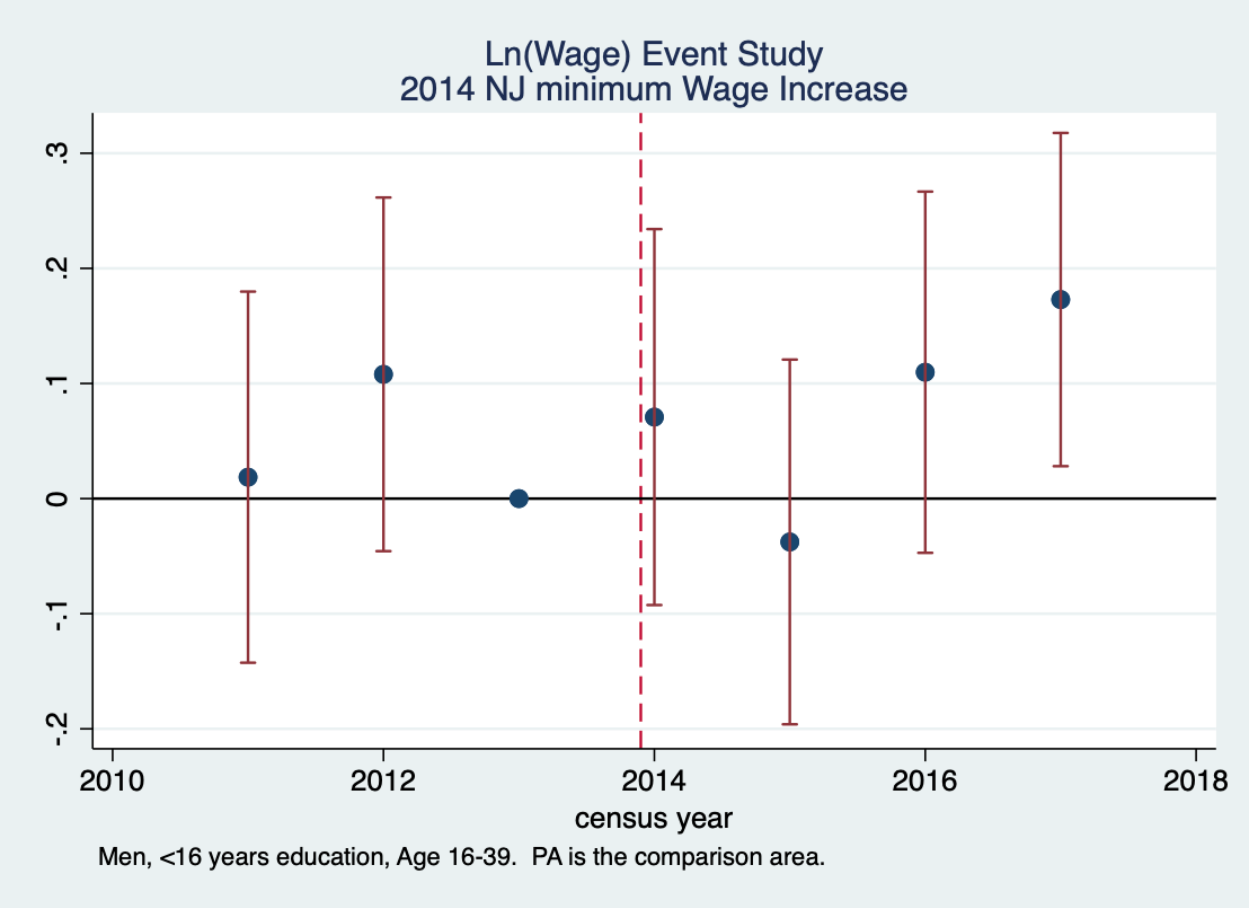
* Step (3): collapse these to the year level
collapse (mean) slope se, by(year)

* Step (4) construct the confidence interval
gen upper = slope + 1.96*se
gen lower = slope - 1.96*se

* Step (5) plot them
gr two (scatter slope year, yline(0, lc(black))) ///
(rcap upper lower year, xline(2013.9, lp(dash))), legend(off) ///
title("Ln(Wage) Event Study" "2014 NJ minimum Wage Increase", size(medium) ///
note("Men, <16 years education, Age 16-39.  PA is the comparison area.")
```

Notice the syntax of the graph command, which I did a little differently this time

- I used a scatter plot instead of a line graph. This is just my preference for this differenced type of event study. A line graph would also be fine
- I added a reference horizontal black line at the origin; this is helpful to visualize if there are pre-trends "`, yline(0, lc(black))`".
- I turned off the legend.
- I split the title across two lines
- Here's what the resulting graph looks like:



Each one of those points represents the difference between NJ and PA in each year, along with a confidence interval on that difference. It is literally the difference between the two lines in the previous graph, with the 2013 difference subtracted out from all years. Thus, it is a bunch of year-specific difference in differences. And you can see there are no significant pretrends (but there is not much significant change after either, until 2017!)

More than one outcome. You could also do this separately for more than one outcome variable. You'd have to run a separate regression for that other outcome variable and create separately named slope and se variables for that outcome, add those to the collapse command, and construct separate upper and lower variables for it. The _b[*variable*] and

_se[*variable*] only refers back to the last regression you ran, kind of like outreg2. To be explicit, here is some code for doing both wages and employment:

```
* step (0a): run the fully interacted regression above, for wages
reg lnrwage yrdum* nj njXyrdum* [w=perwt], robust

* step (1a): create empty variables for the slope and standard error
gen slope = 0
gen se = .

*  Step (2a): copy the slopes and standard errors of each year into these
variables year-by-year and then
foreach y in 2011 2012 2014 2015 2016 2017 {
  replace slope = _b[njXyrdum`y'] if year==`y'
  replace se = _se[njXyrdum`y'] if year==`y'
}

* step (0b): run the fully interacted regression above, for employment
reg employed yrdum* nj njXyrdum* [w=perwt], robust

* step (1b): create empty variables for the slope and standard error
gen slope_emp = 0
gen se_emp = .

*  Step (2b): copy the slopes and standard errors of each year into these
variables year-by-year and then
foreach y in 2011 2012 2014 2015 2016 2017 {
  replace slope_emp = _b[njXyrdum`y'] if year==`y'
  replace se_emp = _se[njXyrdum`y'] if year==`y'
}

* Step (3): collapse these to the year level
collapse (mean) slope se slope_emp se_emp, by(year)

* Step (4) construct the confidence intervals
gen upper = slope + 1.96*se
gen lower = slope - 1.96*se

gen upper_emp = slope_emp + 1.96*se_emp
gen lower_emp = slope_emp - 1.96*se_emp

* step (5) make graphs….
```

Part 4:
Final, Complete Draft

Research Checklist

This is a checklist for writing what I consider a good research paper (in this class or others). If you follow this list, and also avoid saying wrong statements, you will get a good grade on your project. (For additional details, see my lecture notes on research projects) Creativity and originality can help, too, but the key thing for now is to get the basics right:[1]

Paper Structure

- **All papers should include the following sections in this order: (1) Abstract, (2) Introduction, (3) Data and/or Methods (4) Results (5) Discussion and/or Conclusion.**
    - It is also possible to divide these into subsections, or use different titles for the section than these, as long as they are clear.
        - The "and/or" means that you could have one section or two. See more below.
    - If you want to deviate substantially from this traditional structure, please check with me first. Almost always, I will tell you that you should stick to the traditional structure, though some interesting deviations may be possible in very rare cases to better fit with a given analytical approach.
    - You can, however, add sections to this list. A "policy background" section before the methods section (for example) could be helpful in some cases.
    - A literature review section is also not necessary but could be part of the introduction or a separate (early) section.

Let us now consider each of these sections in turn.

1. Include an **abstract** at the beginning of your paper. You will turn in a first draft of the abstract early on in the process. The final version will
    - Will describe what you are estimating (for example, "the impact of NJ's 2014 minimum wage increase on employment)
    - Basic findings
    - Should be short 200-300 words.

2. An **introduction** also has a particular structure. It has some flexibility, but the cardinal rule is always describe your results (even though you just did in the abstract— say it again!) AND never describe the results before describing what you did!! Below is a typical structure:

[1] Writing a good research paper, like any paper, is an "art," which cannot be fully defined by a checklist. But there are certain good practices that I hoping to get you to adopt first, and from there you can refine your art.

- (Optional) Some kind of opening "motivation" statement for why the research question is interesting, and maybe talk about the policy you are studying.
- Next at least one paragraph describing what you estimated, and how you estimated it (by OLS, or with fixed effects, etc.). **You should also mention the dataset by name here (e.g., ACS)**. THIS PARAGRAPH IS NOT OPTIONAL
  - Ideally, do this entirely in words. It is very unusual to have actual estimation equations in an introduction.
  - I think it is also sometimes useful to say in words (again avoid symbols, or the word "error") what you need to be true in order for your estimates to be causal. If you don't do it here, please say it in the methods section (below).
- NEXT a paragraph or at least a couple of sentences summarizing your findings. THIS PARAGRAPH IS ALSO NOT OPTIONAL.
- Important thing to reiterate: the order matters.
  Never describe your results before describing what you did!!
- Introductions should generally not be long. Keep it under two pages.

3. Describe your **data/methods**. There is a separate handout on that. Highlights include: (a) mention your data by name (again!) (b) describe data construction, but avoid "STATA names" for variables (unless you first define them) (c) have a table of means and refer to it (at least briefly) (d) include an estimation equation (e) describe assumptions necessary for causal inference and (f) describe balance test results (again, at least briefly) and whether the support the assumption. They do not have to be in this order, but make sure you hit them all.

4. **Describe your results**. Again, there is a separate handout on that. Highlights include: (a) Write in natural paragraphs. (Unlike in this handout!) (b) Stay focused on your core results. (c) Interpret the magnitude, not just the statistical significance. (d) Avoid unsupported assertions and false statements (e) no raw STATA output – tables should be "self-contained." See also below for more detail.

5. Have a **discussion and conclusion**
   - Can be one section or two
   - A conclusion on its own need not be belabored; it can just be a quick summary of the paper's findings, like a mini-abstraction
   - The discussion is a good place to synthesize the results speculate beyond the paper's factual findings, such as:
     - What you can conclude about the paper's original motivation, for example…

- What model might explain your findings? For example, if you find that minimum wages raise employment, you could say something like "this is potentially consistent with the labor market not being perfectly competitive."
    - Other things you might like to look at to support this interpretation if you had more time/data.
- How other variables you can't control for might be biasing the results
- Whether, on balance, you think you can treat your estimates as close to causal (or perhaps too large or too small)?
- Implications for policy – but be careful! A research analysis is quite limited. For example, there may be other costs or benefits to be considered when deciding on actual policy, so give caveats saying this if you want to make policy recommendations. (Also, keep in mind that research is to be kept separate from advocacy.). For example, instead of saying "Therefore, the minimum wage should be immediately raised in all 50 states" say "according to these estimates, the cost to workers from raising the minimum wage appears to be small, though there may be other actors, such as employers, that may be worse off" or something like that.

Additional Writing Tips – For both results and discussion

<u>Tell a coherent story! / Quality not quantity</u>

6. Not all of your results need to fit perfectly with your overall story (for example, many people find that their significant results disappear when fixed effects are controlled for) but you should not get too far off track the main theme
7. When you're thinking of adding something new, ask yourself – does this help enhance the overall point of the paper, or is it just making the paper more complicated?
    - This is a judgment call, but here's another way to put it: do not include more results than you can write up effectively. Having the paper be effectively written is more important than having a lot of results.
    - Sometimes it's hard to let go of things you tried that did not work out (I want credit for all the things I tried!) If you are having trouble deleting results from your paper because of this, one approach is to mention the other things you tried but didn't work out.
    - (Aside: even if other things you tried <u>did</u> work out, it is also sometimes OK to briefly describe the results without a table; that is, if the paper's getting too long you can just cut a more marginal table and describe briefly what you found in it.)

Write it in the shortest way that is clear.

- Acknowledge, but do not dwell on, alternative interpretations
- Flowery language does not enhance the paper; it makes it harder to understand. Be as concise as possible.[2]
- A common bad writing technique is to vary the word used to describe some concept, as in "In this paper we examine the impact of binge drinking on employment. Heavy drinking has previously been shown to have negative health consequences…." In these two sentences, it is not clear whether 'binge drinking' and 'heavy drinking' refer to the same or different phenomena. It is not making the writing more interesting to vary the word being used, it is just leading to confusion.

Some additional tips (repeat from previous handouts)

Table Tips

1. Never include raw STATA output in your paper

- Instead, use "outreg2" which constructs professional-style tables. More on this in other handouts.

2. Tables should be self-contained
   - That is, there should be sufficient notes and or headers to figure out what is in the regression without having to read the paper. Even with the "outreg2" command output, you may need to further edit your excel spreadsheet to make the final version of the table clear. **I recommend "save as…" and changing the file type to an "Excel Workbook" if you do this, otherwise you may accidentally overwrite your edits when you re-run your STATA program.**
3. Start simple then add complexity. If you are including specifications with a bunch of controls, first show specification without controls.
   - Or, to take more unlikely examples, if you are estimating by IV or logit, first estimate by OLS
   - Also, try to keep the sample as consistent as possible across specification, or else provide a "bridge" between specifications that shows the impact of changing the sample.

Writing

4. Write in natural paragraphs. (Unlike in this handout!) bullet points and checklist style of writing should be kept to a minimum.

[2] Note that this is one place where economics writing may differ significantly from other fields, such as creative writing.

5. Stay focused on your core results. You do not need to discuss the coefficient on every control, etc. Instead, all the writeup should be focused on contributing to the main narrative of the paper.
6. **All** tables that you use should be referred to by their table number somewhere in the text, and **tables should appear in the order they are referred to in the text.**
7. Interpret the magnitude, not just the statistical significance. Statistical significance is not a goal: your project is interesting regardless of what you find.
    - If it is insignificant, and you want to say something further, it is also good to describe the confidence interval.
8. Avoid unsupported assertions!
    - If you are a little unsure, ask yourself: do I need to say this? **It is better to say nothing than to say something wrong.**

Graphs

9. All papers should include at least one graph that is relevant to the estimates. In most cases, this will likely be an "event study" type of graph, showing how the outcome changed in the treatment and comparison areas (groups) both before and after the treatment.
10. Like tables, graphs should be readable. Graph options like title("…") and note("…") and similar xtitle and ytitle options should be used. Again, use the STATA help menus.

Score sheet for group _____

Aspect:	**Max:**	**Score:**
Paper has an abstract	4 points	
Tables included		
Table of Means	4 points	
Balance Test (can be part of Table of means)	4 points	
Controls for year (e.g. "post"), std errors appropriately calculated	4 points	
"Introduction": clearly described, well-motivated topic (can also include the literature review, if there is one, in this)	4 points	
Introduction: quality of explanation of approach	4 points	
Introduction: quality of explanation of findings. **Should include magnitude of main estimates**.	4 points	
Data: look for description of dataset, description of variables, and explanations that are clear.	8 points	
Methods: clear explanation of regression methods used, **including:** • **Estimation equation** (may appear elsewhere in text) • **Assumptions needed for causal inference** (in words)	8 points	
Methods: original/creative/well rationalized methods	8 points	
Note – quality of balance test could influence this score, too		
Results: clear explanation of findings including magnitude of coefficients; clear, appropriately formatted tables: • **Up to -5 points for tables lacking sufficient notes (e.g., describing how standard errors were calculated)**. • Make sure there is an "interpret the slope" sentence somewhere in the text • **Make sure there is at least one graph (-10 if none)**	20 points	
Discussion/Conclusion: thoughtful and correct interpretation of results. Generic but correct interpretation = 8 points. More for more advanced interpretations, including an intelligent discussion of omitted variables bias or thoughtful implications for future research	16 points	
Turned in log files, which are well documented	8 Points	
TOTAL:	100 Points	
Penalties: Subtract Number of days late *squared*		

Note: This is a rough guide. Evaluations will also reflect an overall assessment of the paper as well as the size of the group. In **extreme** cases I reserve the right to give different grades to individual group members. Please work together!